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Abstract

Background: While respiratory conditions are known to increase the risk of psychiatric comorbidities, there remains a significant knowledge gap regarding the specific longitudinal trajectories of incident depression when comparing distinct phenotypes like asthma, COPD, and asthma-COPD overlap (ACO).
Methods: Using CPCSSN data, a data wrangling pipeline was developed to digest raw EHR data, standardize variables, and isolate a cohort with new-onset depression by excluding prior mental health records. Subsequently, time-to-depression was evaluated using Kaplan-Meier and log-rank tests, and a 10% change-in-estimate confounder analysis was performed.
Results: Kaplan-Meier and log-rank analyses revealed overlapping survival curves with no statistically significant differences in time-to-depression among the Asthma, COPD and ACO groups, while confounder analysis identified age as the meaningful variable requiring adjustment (-13.8% change-in-estimate).
Conclusions: The incidence of new-onset depression appears similar across these distinct respiratory conditions, though age remains a critical confounder that must be accounted for in future predictive models.

Introduction

- **Significant Public Health Impact:** Chronic obstructive pulmonary disease (COPD) and asthma represent a substantial and growing burden both in Canada and worldwide¹.
- **High Prevalence:** According to 2018 Canadian statistics, 3.8 million individuals suffer from asthma and 2.0 million from COPD².
- **Psychiatric Comorbidities:** Beyond the physiological impact on respiratory function, both respiratory conditions are frequently comorbid with psychiatric disorders, notably depression^{3,4,5,6}.
- **Evidence on Phenotype Risks:** Recent evidence from a nationwide retrospective cohort study in Korea demonstrates that the risk of mood disorders varies by respiratory phenotype; notably, patients with Asthma-COPD Overlap (ACO) experience a significantly higher incidence of depression (aHR 1.10) compared to those with COPD alone⁷.
- **Impact of Confounding Covariates:** The link between respiratory diseases and depression is confounded by systemic and genetic factors. Adjusting for age, sex, metabolic markers such as HbA1c⁸ and glucose⁹, and a family history of depression^{10, 11} is essential to determine the true incidence of depression.
- Despite established depression risks in ACO versus COPD, the relative risks among asthma-only, COPD-only, and ACO cohorts remain unknown. This study investigates: **“How does the incidence of new-onset depression differ between patients with asthma, COPD and ACO (Asthma-COPD Overlap)?”**

Module 1: Data Input

- Ingest raw EHR data
- Handle “bad-lines”
- Output machine-readable datasets

Module 2: Data Cleaning

- Remove duplicates and standardize missing values
- Standardize categorical, date, and numeric formats

Module 3: Filter Target Cohorts

- Identify target exposure groups
- Establish index date
- Exclude prior depression
- Extract covariates: comorbidities, biomarkers, and family history

Module 4: Survival Analysis

- Kaplan-Meier survival curves
- Two-sided log-rank tests
- Adjust the hazard ratios for relevant baseline covariates (confounders assessment)

Figure 1. Data Wrangling Flowchart

Methods and Materials

Study Population

- Patients were selected from CPCSSN EMR data with asthma or COPD diagnoses (ICD-9). Individuals with prior depression were excluded to capture incident cases. Patients were grouped into asthma-only, COPD-only, and Asthma-COPD Overlap (ACO) cohorts, with follow-up from first diagnosis (index date) to depression onset or censoring.

Data Wrangling

- Raw EMR data were processed using a structured pipeline, including parsing malformed records, standardizing formats, handling missing values, and removing duplicates. Diagnosis codes were harmonized and aggregated to construct patient-level longitudinal datasets.

Cohorts Construction

- Patients were grouped into exposure cohorts (Asthma, COPD and ACO) based on ICD-9 diagnosis history. Index dates were defined as first diagnosis, excluding prior events to ensure incident risk tracking. Age at index and follow-up duration were computed. Baseline covariates included diabetes status (Glucose/HbA1c), family history of depression, and comorbidity count (excluding primary conditions).

Statistical Analysis

- Kaplan–Meier curves estimated depression-free survival, with log-rank tests comparing groups. Confounders were assessed using change-in-estimate criteria (≥10% considered meaningful).

Results

Kaplan–Meier Analysis

The Kaplan–Meier curves showed a decline in depression-free survival over time across all groups (asthma, COPD, and ACO). The curves largely overlapped, suggesting similar time-to-depression patterns.

Log-Rank Test

Pairwise log-rank tests showed no statistically significant differences between groups.

Confounder Analysis

Using a 10% change-in-estimate criterion, **Age** showed meaningful confounding (−13.80%). Other variables (sex, number of comorbidities, diabetes, family history) showed minimal impact (<10%). Although several variables were statistically significant, only age met the criterion for a meaningful confounder.

Table 1. Log Rank Test result.

Group 1	Group 2	Test Statistic	P-value
Asthma	COPD	0.37468	0.54047
Asthma	ACO	0.03042	0.86154
COPD	ACO	0.06634	0.79674

Table 2. Confounder Analysis Result

Confounder	Percent Change	P-value
Age	-13.80	0.0000
Sex	4.78	0.0000
Number of Comorbidities	1.99	0.0000
Diabetes	0.43	0.0536
Family History	0.26	0.0000

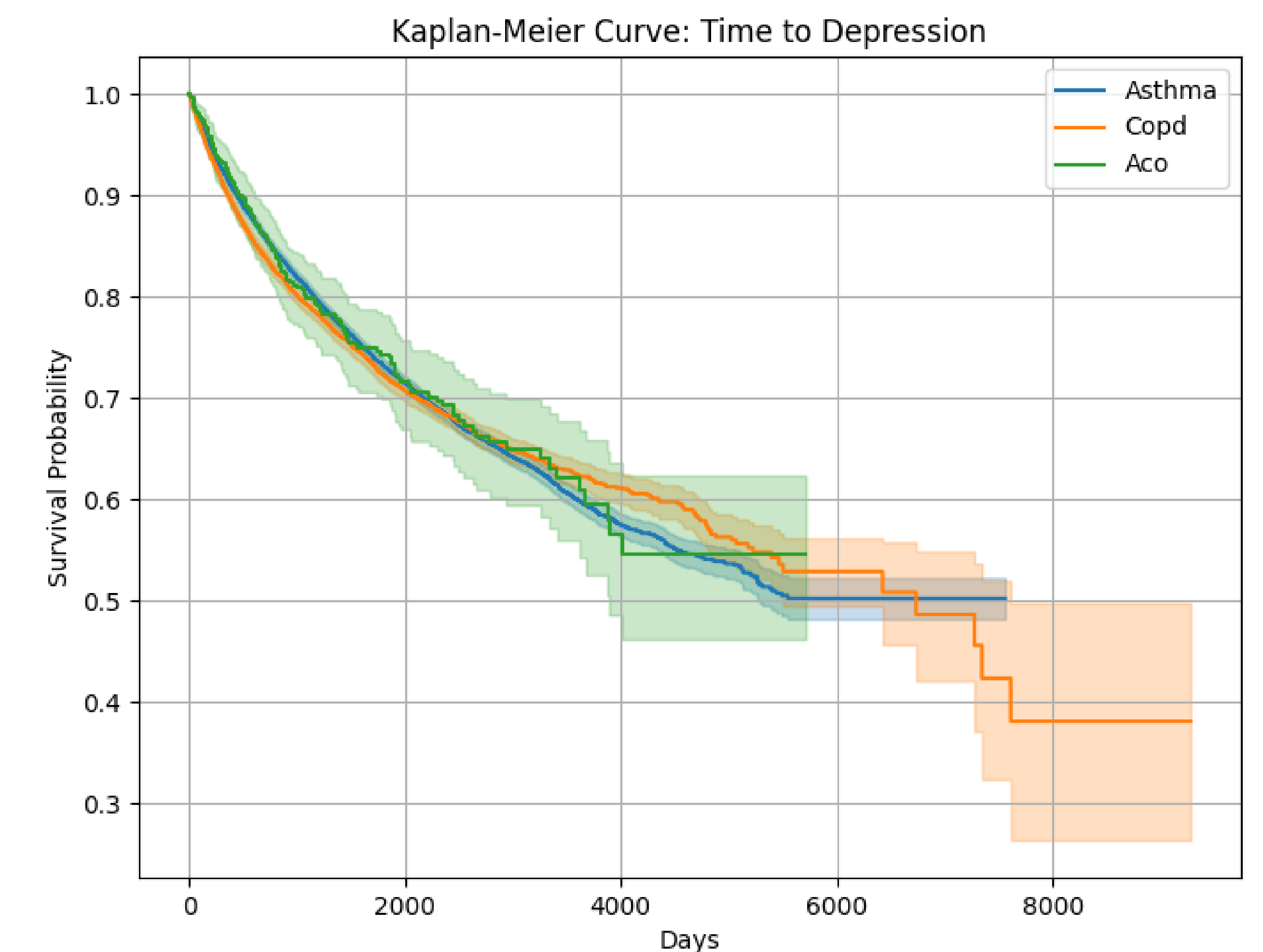


Figure 2. Kaplan Meier Curves for the three groups: Asthma, COPD, and ACO.

Discussion

Depression risk over time appears similar across Asthma, COPD, and ACO groups, with no statistically significant differences observed. Age was identified as a meaningful confounder, while other covariates showed limited impact. These findings suggest that differences in depression incidence may be driven more by demographic factors than disease phenotype. Limitations include potential misclassification in EMR data and residual confounding. Future work should incorporate additional clinical and behavioral variables to improve risk stratification.

Conclusions

- There is no statistically significant difference in time to depression among asthma, COPD, and ACO groups based on Kaplan–Meier analysis and log-rank tests.
- Age was identified as an important confounder and should be adjusted for in subsequent modeling. Overall, the findings suggest that differences in depression incidence across these respiratory conditions are limited after accounting for confounding.

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